AZAD JAMMU & KASHMIR PUBLIC SERVICE COMMISSION

BPS-17 Pure Mathematics Examination

PURE MATHEMATICS

Modern Algebra

Comprehensive Study Plan with MCQs & PSC Tips

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Intermediate Level · Concept + Theorem + MCQ
BPS-17 PSC Reference · 2025–26

Complete Reference —BPS-17 Lecturer Mathematics AJKPSC/FPSC/PPSC/KPSC

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PART I: Group Theory

Groups · Lagrange · Cyclic · Normal Subgroups · Homomorphisms · Isomorphisms · Automorphisms

1. Groups, Subgroups and Lagrange's Theorem

Core Concepts

A **group** $(G, *)$ satisfies: (G1) Closure; (G2) Associativity; (G3) Identity e; (G4) Inverses. An **abelian** (commutative) group additionally satisfies $ab = ba$.

Order of a group $|G|$; **order of an element** $\text{ord}(a) =$ smallest $n > 0$ with $a^n = e$.

Subgroup test: $H \leq G$ iff $H \neq \emptyset$ and $\forall\, a, b \in H : ab^{-1} \in H$.

Cosets: Left coset $aH = \{ah : h \in H\}$; right coset Ha. Cosets partition G.

Theorem: Lagrange's Theorem

If H is a subgroup of a finite group G, then $|H|$ divides $|G|$, and:

$$[G : H] = \frac{|G|}{|H|} = \text{number of distinct left (or right) cosets of } H \text{ in } G.$$

Corollaries: (1) $\text{ord}(a)$ divides $|G|$ for every $a \in G$. (2) $a^{|G|} = e$ for all $a \in G$. (3) Every group of prime order p is cyclic and isomorphic to $\mathbb{Z}_p$.

★ Key Theorems & Formulas ★

Subgroup test: $H \leq G \iff \forall\, a, b \in H : ab^{-1} \in H$

$|G| = |H| \cdot [G : H]$ (Lagrange)

Euler's theorem: $\gcd(a, n) = 1 \Rightarrow a^{\varphi(n)} \equiv 1 \pmod{n}$ (from Lagrange applied to $\mathbb{Z}_n^*$)

Number of subgroups of $\mathbb{Z}_n$ = number of divisors of n

Q1.1. The number of subgroups of $\mathbb{Z}_{12}$ is:

(A) 4
(B) 6
(C) 8
(D) 12

Answer & Full Explanation

Correct Answer: (B) 6

Subgroups of $\mathbb{Z}_n$ correspond bijectively to divisors of n. Divisors of 12: $1, 2, 3, 4, 6, 12$ — that is 6 divisors.

Subgroups: $\{0\} = \langle 6 \rangle$ (order 1 wait — $\langle 6 \rangle = \{0, 6\}$, order 2); more carefully: subgroup of order d is $\langle n/d \rangle$ for each divisor d of n. Orders: 1,2,3,4,6,12 — 6 subgroups total.

Q1.2. If $|G| = 35$, which orders are possible for non-identity elements of G?

- **(A)** 5 or 7 only
- **(B)** $1, 5, 7$ or 35
- **(C)** Any divisor of 35
- **(D)** 5 only

Answer & Full Explanation

Correct Answer: (A) 5 or 7 only

By Lagrange, $\text{ord}(a)$ divides $|G| = 35 = 5 \times 7$. Divisors: $1, 5, 7, 35$. Order 1 is only the identity; order 35 would make G cyclic. By Sylow theory, G must have elements of orders 5 and 7. For PSC purposes: non-identity elements have orders 5 or 7.

Q1.3. A group of prime order p is:

- **(A)** Always non-abelian
- **(B)** Cyclic and isomorphic to $\mathbb{Z}_p$
- **(C)** Has exactly p subgroups
- **(D)** Isomorphic to S_p

Answer & Full Explanation

Correct Answer: (B)

By Lagrange, a group G with $|G| = p$ (prime) has no subgroups of order other than 1 and p. Every non-identity element generates all of G, so G is cyclic. A cyclic group of order p is isomorphic to $\mathbb{Z}_p$.

Q1.4. The left cosets of $H = \{0, 4, 8\}$ in $\mathbb{Z}_{12}$ are:

- **(A)** $H,\ 1 + H,\ 2 + H,\ 3 + H$
- **(B)** $H,\ 1 + H,\ 2 + H$
- **(C)** H only

(D) H and $6 + H$

Answer & Full Explanation

Correct Answer: (A) $H, 1 + H, 2 + H, 3 + H$

$|H| = 3$, $|\mathbb{Z}_{12}| = 12$, so $[\mathbb{Z}_{12} : H] = 12/3 = 4$ cosets. $0 + H = \{0, 4, 8\}$; $1 + H = \{1, 5, 9\}$; $2 + H = \{2, 6, 10\}$; $3 + H = \{3, 7, 11\}$. These four cosets partition $\mathbb{Z}_{12}$.

Q1.5. $|G| = 24$ and $H \leq G$ with $|H| = 6$. Then $[G : H] =$:

(A) 3

(B) 4

(C) 18

(D) 144

Answer & Full Explanation

Correct Answer: (B) 4

$[G : H] = |G|/|H| = 24/6 = 4$. The group G has exactly 4 left cosets of H.

2. Cyclic Groups

Core Concepts

G is **cyclic** if $G = \langle a \rangle = \{a^n : n \in \mathbb{Z}\}$ for some generator a.

Generators of $\mathbb{Z}_n$: $[k]$ generates $\mathbb{Z}_n$ iff $\gcd(k, n) = 1$. Number of generators $= \varphi(n)$.

Order of $[k]$ in $\mathbb{Z}_n$: $\operatorname{ord}([k]) = \dfrac{n}{\gcd(k, n)}$.

Every subgroup of a cyclic group is cyclic.
Every cyclic group is abelian. Converse is false ($V_4 = \mathbb{Z}_2 \times \mathbb{Z}_2$ is abelian but not cyclic).

$\mathbb{Z}_m \times \mathbb{Z}_n$ is cyclic $\iff \gcd(m, n) = 1$.

★ Key Theorems & Formulas ★

$$\operatorname{ord}([k]) \text{ in } \mathbb{Z}_n = \frac{n}{\gcd(k, n)}$$

$$\text{Generators of } \mathbb{Z}_n : \ \{[k] : \gcd(k, n) = 1\}; \ \text{count} = \varphi(n)$$

$$|\langle a \rangle| = \operatorname{ord}(a); \quad \langle a^k \rangle = \langle a^{\gcd(k,n)} \rangle \text{ in cyclic group of order } n$$

$$\mathbb{Z}_{mn} \cong \mathbb{Z}_m \times \mathbb{Z}_n \iff \gcd(m, n) = 1$$

Q2.1. The order of $[8]$ in $\mathbb{Z}_{24}$ is:

- **(A)** 8
- **(B)** 3
- **(C)** 6
- **(D)** 4

Answer & Full Explanation

Correct Answer: (B) 3

$$\operatorname{ord}([8]) = \frac{24}{\gcd(8,24)} = \frac{24}{8} = 3.$$

Check: $8+8+8 = 24 \equiv 0 \pmod{24}$. ✓

Q2.2. Which is **not** a generator of $\mathbb{Z}_{10}$?

- **(A)** $[3]$
- **(B)** $[7]$
- **(C)** $[4]$
- **(D)** $[9]$

Answer & Full Explanation

Correct Answer: (C) $[4]$

Generators of $\mathbb{Z}_{10}$ are elements $[k]$ with $\gcd(k,10) = 1$. $\gcd(4,10) = 2 \neq 1$, so $[4]$ is not a generator. $\gcd(3,10) = \gcd(7,10) = \gcd(9,10) = 1$, so (A),(B),(D) are generators.

Q2.3. $\mathbb{Z}_4 \times \mathbb{Z}_6$ is cyclic because:

- **(A)** $\gcd(4,6) = 2 \neq 1$: it is **not** cyclic
- **(B)** $4+6 = 10$ is composite
- **(C)** Every product of cyclic groups is cyclic
- **(D)** $4 \times 6 = 24$ is even

Answer & Full Explanation

Correct Answer: (A) It is NOT cyclic

$\mathbb{Z}_m \times \mathbb{Z}_n$ is cyclic $\iff$ $\gcd(m,n) = 1$. $\gcd(4,6) = 2 \neq 1$, so $\mathbb{Z}_4 \times \mathbb{Z}_6$ is **not cyclic**. However $\mathbb{Z}_4 \times \mathbb{Z}_3$ is cyclic since $\gcd(4,3) = 1$.

3. Normal Subgroups, Quotient Groups & Homomorphisms

Core Concepts

$N \trianglelefteq G$ (**normal**) $\iff gNg^{-1} = N$ for all $g \in G \iff$ every left coset = right coset.

Quotient group: $G/N = \{gN : g \in G\}$ with $(aN)(bN) = (ab)N$. $|G/N| = |G|/|N| = [G : N]$.

Homomorphism: $\phi : G \to H$ with $\phi(ab) = \phi(a)\phi(b)$.
$\operatorname{Ker}\phi = \{g \in G : \phi(g) = e_H\} \trianglelefteq G$.

Isomorphism: bijective homomorphism. $G \cong H$ means structurally identical.

Theorem: Fundamental Theorem of Homomorphisms (FTH)

If $\phi : G \to H$ is a group homomorphism, then:

$$\operatorname{Ker}\phi \trianglelefteq G \qquad \text{and} \qquad G/\operatorname{Ker}\phi \;\cong\; \operatorname{Im}\phi.$$

First Isomorphism Theorem.

Theorem: Second & Third Isomorphism Theorems

Second: If $H \le G$ and $N \trianglelefteq G$, then $HN \le G$, $H \cap N \trianglelefteq H$, and $\dfrac{H}{H \cap N} \cong \dfrac{HN}{N}$.

Third: If $N \trianglelefteq G$, $K \trianglelefteq G$, $N \le K$, then $K/N \trianglelefteq G/N$ and $\dfrac{G/N}{K/N} \cong \dfrac{G}{K}$.

★ Key Theorems & Formulas ★

$G/\operatorname{Ker}\phi \cong \operatorname{Im}\phi \quad \text{(FTH)}$

$|\operatorname{Ker}\phi| \cdot |\operatorname{Im}\phi| = |G| \quad \text{(for finite } G\text{)}$

$N \trianglelefteq G \iff gNg^{-1} = N \ \forall g \in G \iff$ every subgroup of index 2 is normal

Centre $Z(G) = \{g \in G : gx = xg \ \forall x \in G\} \trianglelefteq G$ always

Q3.1. If $\phi : \mathbb{Z}_{12} \to \mathbb{Z}_4$ is defined by $\phi([n]) = [n \bmod 4]$, then $|\operatorname{Ker}\phi| =$:

- **(A)** 4
- **(B)** 3
- **(C)** 2
- **(D)** 12

Answer & Full Explanation

Correct Answer: (B) 3

$\operatorname{Ker}\phi = \{[n] \in \mathbb{Z}_{12} : n \equiv 0 \pmod 4\} = \{[0], [4], [8]\}$, so $|\operatorname{Ker}\phi| = 3$.
By FTH: $\mathbb{Z}_{12}/\operatorname{Ker}\phi \cong \operatorname{Im}\phi = \mathbb{Z}_4$, so $|\mathbb{Z}_{12}|/|\operatorname{Ker}\phi| = |\mathbb{Z}_4|$, giving $12/|\operatorname{Ker}\phi| = 4$, confirming $|\operatorname{Ker}\phi| = 3$.

Q3.2. Every subgroup of index 2 in a group G is:

(A) Cyclic
(B) Normal in G
(C) Abelian
(D) Isomorphic to $\mathbb{Z}_2$

Answer & Full Explanation

Correct Answer: (B) Normal in G

If $[G : H] = 2$, there are exactly two cosets: H and gH (for $g \notin H$). Since $gH = G \setminus H = Hg$, all left cosets equal right cosets, so $H \trianglelefteq G$.

Q3.3. The quotient group $\mathbb{Z}/4\mathbb{Z}$ is isomorphic to:

(A) $\mathbb{Z}_2 \times \mathbb{Z}_2$
(B) $\mathbb{Z}_4$
(C) $\mathbb{Z}$
(D) S_4

Answer & Full Explanation

Correct Answer: (B) $\mathbb{Z}_4$

The cosets of $4\mathbb{Z}$ in $\mathbb{Z}$ are $\{0+4\mathbb{Z}, 1+4\mathbb{Z}, 2+4\mathbb{Z}, 3+4\mathbb{Z}\}$ — exactly 4 cosets. The natural surjection $\phi : \mathbb{Z} \to \mathbb{Z}_4$, $n \mapsto [n]$, has $\operatorname{Ker}\phi = 4\mathbb{Z}$. By FTH: $\mathbb{Z}/4\mathbb{Z} \cong \mathbb{Z}_4$.

Q3.4. Which of the following is a **normal** subgroup of S_3?

(A) $H = \{e, (12)\}$
(B) $H = \{e, (13)\}$
(C) $A_3 = \{e, (123), (132)\}$
(D) $H = \{e, (23)\}$

Answer & Full Explanation

Correct Answer: (C) A_3

A_3 (the alternating group) is the kernel of the sign homomorphism $\operatorname{sgn} : S_3 \to$

$\{\pm 1\}$. Since every kernel is normal, $A_3 \trianglelefteq S_3$. $|A_3| = 3$ and $[S_3 : A_3] = 2$, confirming normality by the index-2 result. Options (A),(B),(D) are subgroups of order 2 (index 3) and are NOT normal in S_3.

4. Inner Automorphisms, Conjugates & Commutator Subgroup

Core Concepts

Automorphism: An isomorphism $\phi : G \to G$ (self-isomorphism). $\mathrm{Aut}(G)$ is a group under composition.

Inner automorphism by g: $\phi_g(x) = gxg^{-1}$ (conjugation by g). $\mathrm{Inn}(G) = \{\phi_g : g \in G\} \trianglelefteq \mathrm{Aut}(G)$.

$\mathrm{Inn}(G) \cong G/Z(G)$ where $Z(G)$ is the centre.

Conjugate elements: $a, b \in G$ are conjugate if $b = gag^{-1}$ for some g. Conjugacy classes partition G.

Conjugate subgroups: H and gHg^{-1} are conjugate. Normal subgroup = its own conjugate.

Commutator: $[a, b] = a^{-1}b^{-1}ab$. **Commutator subgroup** $G' = [G, G] = \langle [a, b] : a, b \in G\rangle$. G/G' is abelian; G' is the smallest normal subgroup with abelian quotient.

★ Key Theorems & Formulas ★

$$\phi_g(x) = gxg^{-1} \quad \text{(inner automorphism by } g\text{)}$$

$$\mathrm{Inn}(G) \cong G/Z(G) \quad \text{(if } G \text{ is abelian: } Z(G) = G, \mathrm{Inn}(G) = \{e\}\text{)}$$

$$|[a]_\sim| = |G|/|C_G(a)| \quad \text{(size of conjugacy class; } C_G(a) = \text{centraliser of } a\text{)}$$

$$G' = [G, G] \trianglelefteq G; \quad G/G' \text{ is the largest abelian quotient of } G$$

$$\text{Class equation: } |G| = |Z(G)| + \sum_{|[a]|>1} |G|/|C_G(a)|$$

Q4.1. For an abelian group G, the group of inner automorphisms $\mathrm{Inn}(G)$ is:

(A) Isomorphic to G

(B) The trivial group $\{e\}$

(C) $\mathrm{Aut}(G)$

(D) A subgroup of G

Answer & Full Explanation

Correct Answer: (B) The trivial group $\{e\}$

$\text{Inn}(G) \cong G/Z(G)$. If G is abelian, $Z(G) = G$, so $G/Z(G) = G/G = \{eG\}$, the trivial group. Inner automorphisms are $\phi_g(x) = gxg^{-1} = gg^{-1}x = x$ (identity) for abelian G.

Q4.2. The commutator subgroup G' of an abelian group G is:

(A) All of G

(B) The centre $Z(G)$

(C) The trivial subgroup $\{e\}$

(D) A subgroup of index 2

Answer & Full Explanation

Correct Answer: (C) The trivial subgroup $\{e\}$

For abelian G: $[a, b] = a^{-1}b^{-1}ab = a^{-1}ab^{-1}b = e$. So all commutators are trivial, giving $G' = \{e\}$. Equivalently: G itself is abelian, so $G/G' = G/\{e\} \cong G$ is abelian, confirming $G' = \{e\}$.

Q4.3. Two elements a, b of G are conjugate iff:

(A) $ab = ba$

(B) $\exists\, g \in G : b = gag^{-1}$

(C) $a = b$

(D) $a^2 = b^2$

Answer & Full Explanation

Correct Answer: (B) $\exists\, g \in G : b = gag^{-1}$

This is the definition. Conjugacy defines an equivalence relation on G; the equivalence classes are the conjugacy classes. Note: (A) means a and b commute, which is different.

Important PSC Points

Group Theory PSC Must-Know Results:

- Lagrange: $|H|$ divides $|G|$; $\text{ord}(a)$ divides $|G|$.
- Index-2 subgroup is always normal.
- Kernel of any homomorphism is normal.
- FTH: $G/\operatorname{Ker}\phi \cong \operatorname{Im}\phi$.
- Every group of prime order is cyclic.
- $\text{Inn}(G) \cong G/Z(G)$; abelian $G \Rightarrow \text{Inn}(G)$ trivial.
- $G' = \{e\}$ iff G is abelian.
- $\mathbb{Z}_m \times \mathbb{Z}_n$ cyclic $\iff \gcd(m, n) = 1$.

PART II: Ring Theory

Rings · Integral Domains · Fields · Extensions · Finite Fields

5. Rings, Subrings, Integral Domains & Fields

Core Concepts

A **ring** $(R, +, \cdot)$: $(R, +)$ is abelian group; $\cdot$ is associative; distributive laws hold.

Commutative ring: $ab = ba$. **Ring with unity:** has 1_R.

Zero divisor: $a \neq 0$ with $ab = 0$ for some $b \neq 0$.

Integral domain (ID): commutative ring with unity and no zero divisors.

Field: ID where every non-zero element has a multiplicative inverse.

Hierarchy: Field $\subset$ ID $\subset$ Commutative Ring with Unity $\subset$ Ring.

Characteristic of R: smallest $n > 0$ with $n \cdot 1 = 0$ (or 0 if none). Char of a field is 0 or prime.

Theorem: Ideal and Quotient Ring

An **ideal** I of ring R: $(I, +) \leq (R, +)$ and $ri, ir \in I$ for all $r \in R$, $i \in I$.

The **quotient ring** $R/I = \{r + I : r \in R\}$ with $(r + I) + (s + I) = (r + s) + I$ and $(r + I)(s + I) = rs + I$.

R/I is an integral domain $\iff$ I is a **prime ideal**. R/I is a field $\iff$ I is a **maximal ideal**.

★ Key Theorems & Formulas ★

$$\text{Field} \subset \text{ID} \subset \text{Comm. ring (unity)} \subset \text{Ring}$$

$$\mathbb{Z}_n \text{ is a field} \iff n \text{ is prime}; \quad \mathbb{Z}_n \text{ is an ID} \iff n \text{ is prime}$$

$$\text{Every finite ID is a field (Wedderburn)}$$

$$\text{Char of ID is 0 or prime}; \quad \text{Char of } \mathbb{Z}_p = p$$

$$\text{Quotient field of ID } R: \ \{a/b : a, b \in R, b \neq 0\}/\sim$$

Q5.1. In $\mathbb{Z}_{10}$, which element is a zero divisor?

(A) [3]

(B) [7]

(C) [4]

(D) [9]

Answer & Full Explanation

Correct Answer: (C) [4]

$[4] \cdot [5] = [20] = [0]$ in $\mathbb{Z}_{10}$, yet $[4] \neq [0]$ and $[5] \neq [0]$. So [4] is a zero divisor. $\gcd(4, 10) = 2 \neq 1$, confirming [4] is not a unit, hence a zero divisor. $\gcd(3, 10) = \gcd(7, 10) = \gcd(9, 10) = 1$: these are units, not zero divisors.

Q5.2. The characteristic of the ring $\mathbb{Z} \times \mathbb{Z}_6$ is:

(A) 0

(B) 6

(C) 12

(D) 1

Answer & Full Explanation

Correct Answer: (A) 0

The characteristic of a direct product $R_1 \times R_2$ is $\text{lcm}(\text{char}(R_1), \text{char}(R_2))$ (with convention: if either char is 0, the lcm is 0). $\text{char}(\mathbb{Z}) = 0$, $\text{char}(\mathbb{Z}_6) = 6$. $\text{lcm}(0, 6) = 0$.

Q5.3. $\mathbb{Z}_p[x]/\langle f(x)\rangle$ is a field iff:

(A) $\deg f(x) = 2$

(B) $f(x)$ is irreducible over $\mathbb{Z}_p$

(C) p is prime

(D) Both (B) and (C)

Answer & Full Explanation

Correct Answer: (D) Both (B) and (C)

$\mathbb{Z}_p[x]/\langle f(x)\rangle$ is a field $\iff$ $\langle f(x)\rangle$ is a maximal ideal $\iff$ $f(x)$ is irreducible over $\mathbb{Z}_p$. Also, $\mathbb{Z}_p$ must itself be a field (i.e. p prime). Both conditions together are needed.

Q5.4. Which of the following is an integral domain but **not** a field?

(A) $\mathbb{Q}$

(B) $\mathbb{Z}_7$

(C) $\mathbb{Z}$

(D) $\mathbb{R}$

Answer & Full Explanation

Correct Answer: (C) $\mathbb{Z}$ (integers)

$\mathbb{Z}$ is a commutative ring with unity and no zero divisors — hence an integral domain. But $2 \in \mathbb{Z}$ has no multiplicative inverse in $\mathbb{Z}$ ($1/2 \notin \mathbb{Z}$), so $\mathbb{Z}$ is not a field. $\mathbb{Q}$, $\mathbb{Z}_7$, $\mathbb{R}$ are all fields.

6. Field Extensions and Finite Fields

Core Concepts

A **field extension** E/F means F is a subfield of E. The **degree** $[E : F] = \dim_F E$ (as F-vector space).

$\alpha \in E$ is **algebraic** over F if it satisfies a polynomial in $F[x]$. The **minimal polynomial** of α over F is the monic irreducible $p(x) \in F[x]$ with $p(\alpha) = 0$.

$F(\alpha) \cong F[x]/\langle p(x)\rangle$; $[F(\alpha) : F] = \deg p(x)$.

Tower law: $[E : F] = [E : K][K : F]$.

Finite fields (Galois fields): $\mathrm{GF}(p^n) = \mathbb{F}_{p^n}$ exists and is unique (up to isomorphism) for every prime power p^n. $|\mathbb{F}_{p^n}| = p^n$; characteristic $= p$.

★ Key Theorems & Formulas ★

$$[E : F] = \dim_F E; \quad [E : F] = [E : K][K : F] \quad \text{(tower law)}$$
$$F(\alpha) \cong F[x]/\langle p(x)\rangle; \quad [F(\alpha) : F] = \deg p(x)$$
$$|\mathbb{F}_{p^n}| = p^n; \text{ char} = p; \text{ unique up to } \cong$$
$$\text{Subfields of } \mathbb{F}_{p^n} : \mathbb{F}_{p^d} \text{ for each } d \mid n$$
$$\mathbb{F}^*_{p^n} = \mathbb{F}_{p^n} \setminus \{0\} \text{ is cyclic of order } p^n - 1$$

Q6.1. $[\mathbb{Q}(\sqrt{2}) : \mathbb{Q}]$ equals:

- **(A)** 1
- **(B)** 2
- **(C)** 4
- **(D)** ∞

Answer & Full Explanation

Correct Answer: (B) 2

The minimal polynomial of $\sqrt{2}$ over $\mathbb{Q}$ is $x^2 - 2$ (degree 2, irreducible over $\mathbb{Q}$). $[\mathbb{Q}(\sqrt{2}) : \mathbb{Q}] = \deg(x^2 - 2) = 2$. Basis of $\mathbb{Q}(\sqrt{2})$ over $\mathbb{Q}$: $\{1, \sqrt{2}\}$.

Q6.2. The finite field $\mathbb{F}_{2^4} = \mathbb{F}_{16}$ has:

(A) 16 elements, characteristic 4

(B) 16 elements, characteristic 2

(C) 4 elements

(D) 2 elements

Answer & Full Explanation

Correct Answer: (B) 16 elements, characteristic 2

$\mathbb{F}_{p^n}$ has p^n elements and characteristic p. Here $p = 2$, $n = 4$: $|\mathbb{F}_{16}| = 16$, char $= 2$. $\mathbb{F}_{16}$ can be constructed as $\mathbb{F}_2[x]/\langle p(x)\rangle$ where $p(x)$ is irreducible of degree 4 over $\mathbb{F}_2$.

Q6.3. The subfields of $\mathbb{F}_{2^{12}}$ include:

(A) $\mathbb{F}_{2^3}$ and $\mathbb{F}_{2^4}$ only

(B) $\mathbb{F}_{2^d}$ for every divisor d of 12

(C) $\mathbb{F}_{2^6}$ only

(D) All fields $\mathbb{F}_p$ for prime p

Answer & Full Explanation

Correct Answer: (B)

$\mathbb{F}_{p^n}$ contains $\mathbb{F}_{p^d}$ as a subfield $\iff$ $d \mid n$. Divisors of 12: $1, 2, 3, 4, 6, 12$. So subfields: $\mathbb{F}_2, \mathbb{F}_4, \mathbb{F}_8, \mathbb{F}_{16}, \mathbb{F}_{64}, \mathbb{F}_{4096} = \mathbb{F}_{2^{12}}$.

PART III: Vector Spaces & Linear Maps

Subspaces · Bases · Dimension · Linear Transformations · Matrices · Rank & Nullity

7. Vector Spaces, Bases and Dimension

Core Concepts

A **vector space** V over field F: $(V, +)$ abelian group with scalar multiplication satisfying 8 axioms.

Subspace test: $W \leq V$ iff $\mathbf{0} \in W$ and W is closed under $+$ and scalar multiplication.

Span: $\text{span}\{v_1, \ldots, v_k\} = \{c_1v_1 + \cdots + c_kv_k : c_i \in F\}$.

Linear independence: $\{v_1, \ldots, v_k\}$ is LI iff $c_1v_1 + \cdots + c_kv_k = \mathbf{0} \Rightarrow$ all $c_i = 0$.

Basis: LI spanning set. All bases of V have the same cardinality $= \dim V$.

Dimension formula: If $W_1, W_2 \leq V$ (finite-dimensional): $\dim(W_1 + W_2) = \dim W_1 + \dim W_2 - \dim(W_1 \cap W_2)$.

★ Key Theorems & Formulas ★

$$\dim \mathbb{R}^n = n; \ \dim M_{m\times n}(F) = mn; \ \dim P_n(F) = n + 1; \ \dim F[x] = \infty$$

$$\dim(W_1 + W_2) = \dim W_1 + \dim W_2 - \dim(W_1 \cap W_2)$$

$$\text{If } \dim V = n: \text{ any } n \text{ LI vectors form a basis; any spanning set } \geq n \text{ vectors}$$

$$\text{Direct sum: } V = W_1 \oplus W_2 \iff V = W_1 + W_2 \text{ and } W_1 \cap W_2 = \{\mathbf{0}\}$$

Q7.1. The dimension of the solution space (null space) of the system $Ax = 0$ where A is 3×5 with rank $A = 2$ is:

(A) 2

(B) 3

(C) 5

(D) 1

Answer & Full Explanation

Correct Answer: (B) 3

By the Rank-Nullity Theorem: $\text{rank}\, A + \text{nullity}\, A = n$ (number of columns). $2 + \text{nullity}\, A = 5 \Rightarrow \text{nullity}\, A = 3$. The solution space (null space) has dimension

3.

Q7.2. Which set is a subspace of $\mathbb{R}^3$?

- **(A)** $\{(x, y, z) : x + y + z = 1\}$
- **(B)** $\{(x, y, z) : x^2 + y^2 = z^2\}$
- **(C)** $\{(x, y, z) : x + y = 0\}$
- **(D)** $\{(x, y, z) : xy = 0\}$

Answer & Full Explanation

Correct Answer: (C) $\{(x, y, z) : x + y = 0\}$

(C): Contains $\mathbf{0}$ (yes: $0 + 0 = 0$). Closed under addition and scalar mult (linear equation). ✓

(A): Does not contain $\mathbf{0}$ ($0 + 0 + 0 = 0 \neq 1$). NOT a subspace.

(B) and (D): Not closed under addition (nonlinear conditions).

Q7.3. If $V = P_3(\mathbb{R})$ (polynomials of degree ≤ 3) and $W = \{p \in V : p(1) = 0\}$, then $\dim W =$:

- **(A)** 2
- **(B)** 3
- **(C)** 4
- **(D)** 1

Answer & Full Explanation

Correct Answer: (B) 3

$\dim V = 4$ (basis: $\{1, x, x^2, x^3\}$). W is the kernel of the linear functional $\mathrm{ev}_1 : p \mapsto p(1)$. Since ev_1 is surjective onto $\mathbb{R}$, $\dim \mathrm{Ker} = \dim V - 1 = 3$. Explicit basis: $\{x - 1, x^2 - 1, x^3 - 1\}$.

Q7.4. Three vectors $(1, 0, 0)$, $(0, 1, 0)$, $(1, 1, 0)$ in $\mathbb{R}^3$:

- **(A)** Form a basis for $\mathbb{R}^3$
- **(B)** Are linearly dependent
- **(C)** Span $\mathbb{R}^3$
- **(D)** Are linearly independent

Answer & Full Explanation

Correct Answer: (B) Are linearly dependent

$(1, 1, 0) = (1, 0, 0) + (0, 1, 0)$, so the third vector is a linear combination of the first two. They are **linearly dependent**. They span only a 2-dimensional subspace (the xy-plane), not all of $\mathbb{R}^3$.

8. Linear Transformations and Matrices

Core Concepts

$T : V \to W$ is a **linear transformation** if $T(au + bv) = aT(u) + bT(v)$.

$\operatorname{Ker} T = \{v \in V : T(v) = \mathbf{0}\}$ (null space); $\operatorname{Im} T = \{T(v) : v \in V\}$ (range).

Matrix representation: Choose bases β for V, γ for W; column j of $[T]_\beta^\gamma$ is the coordinate vector of $T(v_j)$ in γ.

Matrix algebra: $AB \neq BA$ in general; $(AB)C = A(BC)$; $(AB)^T = B^T A^T$.

Row echelon form (REF): Leading entry of each row is to the right of the one above; rows of zeros at bottom.
Reduced REF: Each leading entry is 1 and is the only nonzero entry in its column.

Theorem: Rank-Nullity Theorem

For $T : V \to W$ with $\dim V = n$ (finite):

$$\dim \operatorname{Ker} T + \dim \operatorname{Im} T = \dim V$$

i.e. $\operatorname{nullity}(T) + \operatorname{rank}(T) = n$. For an $m \times n$ matrix A: $\operatorname{rank}(A) + \operatorname{nullity}(A) = n$.

★ Key Theorems & Formulas ★

$$\operatorname{nullity}(T) + \operatorname{rank}(T) = \dim V = n$$

$$\operatorname{rank}(AB) \leq \min(\operatorname{rank} A, \operatorname{rank} B); \quad \operatorname{rank}(A + B) \leq \operatorname{rank} A + \operatorname{rank} B$$

$$T \text{ injective} \iff \operatorname{Ker} T = \{\mathbf{0}\} \iff \operatorname{rank} T = \dim V$$

$$T \text{ surjective} \iff \operatorname{rank} T = \dim W$$

$$T \text{ isomorphism} \iff \dim V = \dim W \text{ and } T \text{ injective (or surjective)}$$

$$\operatorname{rank}(A^T) = \operatorname{rank}(A); \quad \operatorname{rank}(A) = \text{row rank} = \text{column rank}$$

Q8.1. If $T : \mathbb{R}^4 \to \mathbb{R}^3$ is linear with $\operatorname{rank} T = 3$, then $\dim \operatorname{Ker} T =$:

(A) 0

(B) 1

(C) 3

(D) 4

Answer & Full Explanation

Correct Answer: (B) 1

Rank-Nullity: $\operatorname{nullity} T + \operatorname{rank} T = \dim V = 4$. $\operatorname{nullity} T + 3 = 4 \Rightarrow \operatorname{nullity} T = 1$.

Q8.2. The reduced row echelon form of $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{pmatrix}$ is:

(A) $\begin{pmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$

(B) $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

(C) $\begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix}$

(D) $\begin{pmatrix} 3 & 6 & 9 \\ 2 & 4 & 6 \\ 1 & 2 & 3 \end{pmatrix}$

Answer & Full Explanation

Correct Answer: (A)

$R_2 \leftarrow R_2 - 2R_1$: $(0,0,0)$. $R_3 \leftarrow R_3 - 3R_1$: $(0,0,0)$. RREF: $\begin{pmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$.

rank $= 1$; null space has dimension 2.

Q8.3. The rank of matrix $A = \begin{pmatrix} 1 & 2 & -1 \\ 3 & 6 & -3 \end{pmatrix}$ is:

(A) 1

(B) 2

(C) 0

(D) 3

Answer & Full Explanation

Correct Answer: (A) 1

$R_2 = 3R_1$, so the rows are linearly dependent. After row reduction: $\begin{pmatrix} 1 & 2 & -1 \\ 0 & 0 & 0 \end{pmatrix}$.

$\operatorname{rank} A = 1$ (one pivot row). $\operatorname{nullity} A = 3 - 1 = 2$ (3 columns).

Q8.4. For a linear transformation $T : V \to V$ to be an isomorphism (when $\dim V = n$), it suffices that:

- **(A)** T is injective only
- **(B)** T is surjective only
- **(C)** $\operatorname{rank} T = n$
- **(D)** Any one of (A), (B), or (C)

Answer & Full Explanation

Correct Answer: (D)

For $T : V \to V$ with $\dim V = n$ (finite): injective $\iff$ surjective $\iff$ bijective $\iff$ $\operatorname{rank} T = n$. This equivalence holds precisely when domain and codomain have the same finite dimension. Any one of these conditions implies the others.

PART IV: Systems of Equations & Determinants

Homogeneous · Non-homogeneous · Consistency · Cramer's Rule · Determinant Properties

9. Systems of Linear Equations

Core Concepts

System $Ax = b$ where $A \in M_{m\times n}(F)$, $x \in F^n$, $b \in F^m$.

Homogeneous system $Ax = \mathbf{0}$: Always consistent (trivial solution $x = \mathbf{0}$). Non-trivial solution $\iff$ $\operatorname{rank} A < n$. Solution space is a subspace of F^n of dimension $n - \operatorname{rank} A$.

Non-homogeneous system $Ax = b$: Consistent $\iff$ $b \in$ column space of A $\iff$ $\operatorname{rank}[A|b] = \operatorname{rank} A$.

Unique solution: $\operatorname{rank} A = \operatorname{rank}[A|b] = n$.
Infinitely many: $\operatorname{rank} A = \operatorname{rank}[A|b] < n$.
No solution: $\operatorname{rank} A < \operatorname{rank}[A|b]$.

Theorem: Rouché–Capelli (Kronecker–Capelli) Theorem

The system $Ax = b$ is consistent $\iff$ $\operatorname{rank}(A) = \operatorname{rank}([A \mid b])$.

When consistent: if $\operatorname{rank} A = n$ (number of unknowns) then the solution is **unique**; if $\operatorname{rank} A < n$ then there are **infinitely many** solutions (parameterised by $n - \operatorname{rank} A$ free variables).

★ Key Theorems & Formulas ★

$$Ax = \mathbf{0} : \ \dim(\text{soln space}) = n - \operatorname{rank} A$$

$$Ax = b \text{ consistent } \iff \operatorname{rank} A = \operatorname{rank}[A \mid b]$$

$$Ax = b \text{ has unique soln } \iff \operatorname{rank} A = \operatorname{rank}[A \mid b] = n$$

$$\text{General solution: } x = x_p + x_h \ (x_p = \text{particular}, \ x_h \in \operatorname{Ker} A)$$

$$\text{Cramer's rule: } x_i = \det(A_i)/\det(A) \ (\det A \neq 0)$$

Q9.1. The system $x + y + z = 0, \ 2x + 3y + z = 0, \ x + 2y = 0$ has:

(A) Only the trivial solution

(B) Exactly one non-trivial solution

(C) Infinitely many solutions

(D) No solution

Answer & Full Explanation

Correct Answer: (C) Infinitely many solutions

Homogeneous system $Ax = \mathbf{0}$. Row-reduce A: $\begin{pmatrix} 1 & 1 & 1 \\ 2 & 3 & 1 \\ 1 & 2 & 0 \end{pmatrix} \to \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & -1 \\ 0 & 1 & -1 \end{pmatrix} \to$ $\begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix}$. rank $A = 2 < n = 3$, so dimension of solution space $= 3 - 2 = 1$.
Infinitely many solutions (one free variable).

Q9.2. For the system $Ax = b$ with augmented matrix rank: rank $A = 2$, rank$[A|b] = 3$, $n = 4$:

(A) Unique solution

(B) Infinitely many solutions

(C) No solution

(D) Trivial solution only

Answer & Full Explanation

Correct Answer: (C) No solution

By Rouché–Capelli: $Ax = b$ is consistent $\iff$ rank A = rank$[A|b]$. Here rank $A = 2 \neq 3 =$ rank$[A|b]$, so the system is **inconsistent** (no solution).

Q9.3. The solution to $x + 2y = 5,\ 3x + 4y = 11$ using Cramer's rule has $x =$:

(A) 1

(B) 2

(C) -1

(D) 3

Answer & Full Explanation

Correct Answer: (A) $x = 1$

$\det A = \begin{vmatrix} 1 & 2 \\ 3 & 4 \end{vmatrix} = 4 - 6 = -2.$

$\det A_1 = \begin{vmatrix} 5 & 2 \\ 11 & 4 \end{vmatrix} = 20 - 22 = -2.$ $x = \det A_1 / \det A = (-2)/(-2) = 1.$

Check: $1 + 2y = 5 \Rightarrow y = 2$; $3(1) + 4(2) = 11$. ✓

Q9.4. A homogeneous system $Ax = 0$ with $A \in M_{3\times 4}$ always has:

- **(A)** Only the trivial solution
- **(B)** Non-trivial solutions
- **(C)** Exactly 4 solutions
- **(D)** No solution

Answer & Full Explanation

Correct Answer: (B) Non-trivial solutions

A is 3×4, so $\operatorname{rank} A \le 3 < 4 = n$. Dimension of null space $= n - \operatorname{rank} A \ge 1 > 0$. So there exist non-trivial solutions.

General principle: If $n > m$ (more unknowns than equations), a homogeneous system always has non-trivial solutions.

10. Properties of Determinants

Core Concepts

The **determinant** $\det : M_{n\times n}(F) \to F$ is the unique multilinear, alternating function with $\det I = 1$.

Expansion by row i: $\det A = \sum_{j=1}^{n} (-1)^{i+j} a_{ij} M_{ij}$ (cofactor expansion).

Key properties:

- **P1.** $\det A = 0 \iff A$ is singular ($\iff$ rows/columns are linearly dependent).
- **P2.** $\det(AB) = (\det A)(\det B)$.
- **P3.** $\det(A^T) = \det A$.
- **P4.** Row swap: det changes sign. Row scale by k: det multiplied by k.
- **P5.** Adding multiple of one row to another: det unchanged.
- **P6.** $\det(kA) = k^n \det A$ (for $n\times n$ matrix A).
- **P7.** $\det(A^{-1}) = 1/\det A$ (if A invertible).

★ Key Theorems & Formulas ★

$$\det\begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc; \quad \det\begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} = a(ei - fh) - b(di - fg) + c(dh - eg)$$

$$\det(AB) = \det A \cdot \det B; \quad \det(A^T) = \det A; \quad \det(A^{-1}) = 1/\det A$$

$$\det(kA) = k^n \det A \quad (n\times n); \quad \det A = 0 \iff \text{singular}$$

$$A^{-1} = \frac{1}{\det A}\operatorname{adj}(A); \quad \operatorname{adj}(A)_{ij} = (-1)^{i+j} M_{ji}$$

$|\det A| =$ volume of parallelepiped formed by rows (or columns)

Q10.1. If $\det A = 3$ and A is 3×3, then $\det(2A) =$:

(A) 6

(B) 12

(C) 24

(D) 9

Answer & Full Explanation

Correct Answer: (C) 24

$\det(kA) = k^n \det A$. For $n = 3$, $k = 2$: $\det(2A) = 2^3 \cdot \det A = 8 \times 3 = 24$.

Common trap: Many students write $2 \cdot \det A = 6$. The scalar k multiplies **each row**, so det is multiplied by k^n.

Q10.2. If two rows of matrix A are identical, then $\det A =$:

(A) 1

(B) $2 \det A$

(C) 0

(D) Cannot be determined

Answer & Full Explanation

Correct Answer: (C) 0

If two rows are identical, swapping them gives the same matrix, but swapping rows negates det. So $\det A = -\det A$, implying $2 \det A = 0$, i.e. $\det A = 0$.

Equivalently: identical rows are linearly dependent, so A is singular, giving $\det A = 0$.

Q10.3. $\det(AB) = \det A \cdot \det B$ requires:

(A) A and B are square of the same size

(B) A and B are invertible

(C) $\det A \neq 0$

(D) $A = B^T$

Answer & Full Explanation

Correct Answer: (A)

The product formula $\det(AB) = \det A \cdot \det B$ holds for any two $n \times n$ square matrices of the same size, regardless of whether they are invertible.

Q10.4. The determinant of a triangular matrix equals:

(A) Sum of diagonal entries
(B) Product of diagonal entries
(C) Sum of all entries
(D) Trace of the matrix

Answer & Full Explanation

Correct Answer: (B) Product of diagonal entries

For upper or lower triangular matrices, cofactor expansion reduces to the product of diagonal entries: $\det A = a_{11}a_{22}\cdots a_{nn}$. This also shows that a triangular matrix is invertible $\iff$ all diagonal entries are nonzero.

Q10.5. $\det\begin{pmatrix}1 & 2 & 3\\ 0 & 4 & 5\\ 0 & 0 & 6\end{pmatrix} =$

(A) 10
(B) 15
(C) 24
(D) 6

Answer & Full Explanation

Correct Answer: (C) 24

Upper triangular matrix: $\det = 1 \times 4 \times 6 = 24$.

11. Mixed PSC Past-Paper Style MCQs

All Parts Combined — BPS-17/PPSC/FPSC/CSS Standard

Q11.1. The centre $Z(G)$ of any group G is:

(A) Always trivial
(B) Always equal to G
(C) Always a normal subgroup of G
(D) A normal subgroup only if G is abelian

Answer & Full Explanation

Correct Answer: (C)

$Z(G) = \{g \in G : gx = xg \ \forall x \in G\}$ is always normal in G. Proof: For $z \in Z(G)$

and $g \in G$: $gzg^{-1} = gg^{-1}z = z \in Z(G)$. If G abelian: $Z(G) = G$. If G trivial: $Z(G) = \{e\}$. But $Z(G) \trianglelefteq G$ always.

Q11.2. The First Isomorphism Theorem for rings states: if $\phi : R \to S$ is a ring homomorphism, then:

(A) $R \cong S$

(B) $R/\operatorname{Ker}\phi \cong \operatorname{Im}\phi$

(C) $\operatorname{Ker}\phi \cong \operatorname{Im}\phi$

(D) $R \cong \operatorname{Ker}\phi \times \operatorname{Im}\phi$

Answer & Full Explanation

Correct Answer: (B) $R/\operatorname{Ker}\phi \cong \operatorname{Im}\phi$

Exactly analogous to the group FTH. $\operatorname{Ker}\phi = \{r \in R : \phi(r) = 0_S\}$ is an ideal of R, and the induced map $\bar{\phi} : R/\operatorname{Ker}\phi \to \operatorname{Im}\phi$ is a ring isomorphism.

Q11.3. If A is an $n \times n$ invertible matrix, then $\det(A^{-1}) =$:

(A) $-\det A$

(B) $\det A$

(C) $\dfrac{1}{\det A}$

(D) $(\det A)^{-1} = \dfrac{1}{\det A}$ (same as C)

Answer & Full Explanation

Correct Answer: (D) $1/\det A$

$A \cdot A^{-1} = I \Rightarrow \det(A)\det(A^{-1}) = \det(I) = 1 \Rightarrow \det(A^{-1}) = 1/\det(A)$. Options (C) and (D) express the same thing.

Q11.4. A linear transformation $T : \mathbb{R}^3 \to \mathbb{R}^3$ has rank 2. Then:

(A) T is invertible

(B) $\dim \operatorname{Ker} T = 2$

(C) $\dim \operatorname{Ker} T = 1$

(D) T is the zero map

Answer & Full Explanation

Correct Answer: (C) $\dim \operatorname{Ker} T = 1$

Rank-Nullity: $\operatorname{rank} T + \operatorname{nullity} T = 3$. $2 + \operatorname{nullity} T = 3 \Rightarrow \operatorname{nullity} T = 1 =$

$\dim \operatorname{Ker} T$. T is not invertible (rank $\neq 3$) and not zero (rank $\neq 0$).

Q11.5. $\det A = 0$ implies:

(A) $A = 0$ (zero matrix)

(B) A is not invertible

(C) $\operatorname{tr} A = 0$

(D) All eigenvalues are zero

Answer & Full Explanation

Correct Answer: (B) A is not invertible (singular)

$\det A = 0 \iff A$ is singular $\iff A^{-1}$ does not exist $\iff$ rows (or columns) are linearly dependent $\iff Ax = 0$ has non-trivial solutions. (A) is false: $A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ has $\det = 0$ but $A \neq 0$.

12. Quick-Revision Master Summary

Modern Algebra — Complete Reference (Zamir Hussain)

Lagrange's theorem	$\lvert H\rvert$ divides $\lvert G\rvert$; $[G : H] = \lvert G\rvert/\lvert H\rvert$; $\operatorname{ord}(a)$ divides $\lvert G\rvert$
Index-2 normal	Every subgroup of index 2 is normal
Prime order	Group of prime order p is cyclic $\cong \mathbb{Z}_p$
FTH (groups)	$G/\operatorname{Ker}\phi \cong \operatorname{Im}\phi$
Cyclic order	$\operatorname{ord}([k])$ in $\mathbb{Z}_n$ is $n/\gcd(k,n)$
$\mathbb{Z}_m \times \mathbb{Z}_n$ **cyclic**	iff $\gcd(m,n) = 1$
$\operatorname{Inn}(G)$	$\cong G/Z(G)$; trivial if G abelian
Commutator G'	$= \{e\}$ iff G abelian; G/G' = largest abelian quotient
$\mathbb{Z}_n$ **field**	iff n is prime (also char $= n$)
Every finite ID	is a field (Wedderburn's theorem)
Field extension	$[F(\alpha) : F] = \deg(\text{min. poly. of } \alpha)$; tower law
Finite field $\mathbb{F}_{p^n}$	exists uniquely; $(\mathbb{F}_{p^n})^*$ is cyclic of order $p^n - 1$
Rank-Nullity	$\operatorname{rank} T + \operatorname{nullity} T = \dim V$
$Ax = 0$ **soln space**	dimension $= n - \operatorname{rank} A$
$Ax = b$ **consistency**	$\operatorname{rank} A = \operatorname{rank}[A\|b]$; unique iff also $= n$
$\det(kA)$	$= k^n \det A$ for $n \times n$
$\det(AB)$	$= \det A \cdot \det B$ (square, same size)
	= product of diagonal entries

PSC Exam Tip

Top 15 BPS-17/PSC Strategies — Modern Algebra (Zamir Hussain):

1. **Lagrange**: $|H|$ divides $|G|$ — all possible subgroup orders must divide $|G|$.
2. **Index-2 subgroup is always normal** — most tested normality criterion.
3. **FTH**: $G/\operatorname{Ker}\phi \cong \operatorname{Im}\phi$ — connect kernel size to image via $|G|/|\operatorname{Ker}| = |\operatorname{Im}|$.
4. **Prime-order group**: always cyclic $\cong \mathbb{Z}_p$ — no proper non-trivial subgroups.
5. **Order of $[k]$ in $\mathbb{Z}_n$**: $n/\gcd(k,n)$ — most-tested single formula.
6. **Generators of $\mathbb{Z}_n$**: elements coprime to n; count $= \varphi(n)$.
7. **$\mathbb{Z}_n$ is field iff n is prime** — and ID under same condition.
8. **Characteristic**: of an integral domain is 0 or prime.
9. **Rank-Nullity**: $\operatorname{rank} + \operatorname{nullity} = n$ — link between injective/surjective properties.
10. $Ax = b$ **consistency**: $\operatorname{rank} A = \operatorname{rank}[A|b]$ — Rouché-Capelli.
11. $\det(kA) = k^n \det A$: scaling a matrix scales det by k^n, not k.
12. **Triangular determinant = product of diagonals** — fastest det computation.
13. $\det A = 0 \iff$ **singular** — links determinant to invertibility and LI of rows.
14. $[\mathbb{Q}(\sqrt{d}) : \mathbb{Q}] = 2$ for non-square d; $[F(\alpha) : F] = \deg p_\alpha$.
15. **Subspace test**: $\mathbf{0} \in W$, closed under $+$ and scalar multiplication (linear condition required — not $x^2 + y^2 = 1$, etc.).

Exam Practice Resource

For further exam practice, the following resource provides a comprehensive collection of solved MCQs:
Access the material here

Zamir Hussain

BPS-17 PSC Pure Mathematics — Modern Algebra

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